REVIEW
MAS291



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CHAPTER 1,2

Data that describe a characteristic about a population is known as a _____.

A. statistic

B. None of the other choices is true

C. survey

D. sample

E. parameter

A card is drawn from a standard deck of 52 playing cards. Find the probabity that the card is an ace or a king.

A. 1/13

B. 2/13

C. 4/13

D. None of the other choices is correct

E. 8/13

$$P(A) = \frac{8}{52} = \frac{2}{13}$$

A pair of dice is thrown twice. What is the probability of getting totals of 7 and 11

A. 1/54

B. 1/24

C. None of the other choices is correct

D. 1/18

E. 1/14

$$2P(S=7)*P(S=11)$$

$$=2*\frac{6}{36}*\frac{2}{36}=\frac{1}{54}$$

A group of students were asked if they carry a credit card. The responses are listed in the table. If a student is selected at random, find the probability that he or she owns a credit card given that the student is a freshman. Round your answer to three decimal places.

Class	Credit Card Carrier	Not a Credit Card Carrier	Total
Freshman	24	36	60
Sophomore	37	3	40
Total	61	39	100

A. 0.600 B. 0.240 C. 0.400

D. None of the other choices is correct E. 0.393

P= 24/60

Given two events A and B with P(A) = 0.4, P(B) = 0.5. Find P(A U B).

A. 0.9, if A and B are independent.

B. All of the other choices are incorrect

C. 0.7, if A and B are disjoint.

D. 0.7, if A and B are independent.

E. 0.2, if A and B are independent.

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$
$$= 0.4 + 0.5 - 0.4 \times 0.5 = 0.7$$

Two white mice mate. The male has both a white and a black fur-color gene. The female has only white fur-color genes. The fur color of the offspring depends on the pairs of fur-color genes that they receive. Assume that neither the white nor the black gene dominates. List the possible outcomes. W = white and B = black. Create the sample space of possible outcomes.

A. WW, WW
B. None of the other choices is correct
C. WB, BW
D. WW, BB
E. WW, BW

Flip a coin three times, create the sample space of possible outcomes (H: head, T: tail)

a. HHH HTT HTH TTT HTT THH HHT THT
b. HHH HTT TTH TTT HTH THH HHT THT
c. HHH HTT HTH TTT HTT THT
d. HHH TTT HTT THH HHT THT

The distribution of B.A. degrees conferred by a local college is listed below, by major.

Major	Frequency
Engineering	868
English	2073
Mathematics	2164
Chemistry	318
Physics	856
Liberal Arts	1358
Business	1676

What is the probability that a randomly selected degree is not in Chemistry?

According to a survey of American households, the probability that the residents own 2 cars if annual household income is over $30000 is 70%. Of the household surveyed, 50% had incomes over $30000 and 70% had 2 cars. The probability that the residents of a household own 2 cars and have an income over $30000 a year is:

a. 0.22 b. 0.35 c. 0.48 d. 0.18

P(own 2 cars and income > 30,000)

= P(income > 30,000)P(own 2 cars | income > 30,000)

=0.5(0.7)

= 0.35

A bin contains 15 defective (that immediately fail when put in use), 20 partially defective (that fail after a couple of hours of use), and 30 acceptable transistors. A transistor is chosen at random from the bin and put into use. If it does not immediately fail, what is the probability it is acceptable?

$P = 30/50$

Given that events A and B are mutually exclusive and P(A) = 0.2 and P(B) = 0.7, are A and B independent?

P(A and B) = 0

P(A)P(B) = 0.2*0.7

In orange county, 51% of the adults are males. One adult is randomly selected for a survey involving credit card usage. It is later learned that the selected survey subject was smoking a cigar. Also, 7.5% of males smoke cigars, whereas 1.9% of females smoke cigars. Use this additional information to find the probability that the selected subject is a male.

$$P(B) = P(A)P(B|A) + P(A')P(B|A')$$
$$= 0.51*0.075 + 0.49*0.019 = 0.04756$$

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} = \frac{0.51*0.075}{0.04756} = 0.804$$

CHAPTER 3

A basketball player is asked to shot free throws in sets of four. The player shoots 100 sets of 4 free throws. The probability distribution for making a particular number of free throws is given below. Determine the variance for this discrete probability distribution.

X	P(X)
0	0.12
1	0.17
2	0.22
3	0.17
4	0.32

The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent. What is the probability that the first successful alignment requires exactly four trials?

A. 0.6756

B. 0.1262

C. None of the other choices is correct

D. 0.0064

E. 0.4332

$$P(X = 4) = 0.2^3 0.8 = 0.0064$$

Suppose that X has a discrete uniform distribution on the integers 1 to 9. Which of the followings are true?

(i) P (X > 5) = 4/9

(ii) E (3X) = 15

A. (i) only

B. (ii) only

C. Both (i) and (ii)

D. None of the other choices is true

$$P(X>5)=\frac{9-5}{9-1+1}=\frac{4}{9}$$

$$E(3X)=3\frac{9+1}{2}=15$$

Find the mean for the binomial distribution which has the stated values of n=20 and p=0.4. Round answer to the nearest tenth.

A. 12.0
B. 11.5
C. 12.7
D. None of the other choices is correct
E. 8.0

The random variable X represents the number of credit cards that adults have along with the corresponding probabilities. Find the mean and standard deviation.

X	P(X)
0	0.39
1	0.15
2	0.22
3	0.17
4	0.07

A. mean: 1.39; standard deviation: 0.64
B. mean: 1.38; standard deviation: 1.33
C. mean: 1.18; standard deviation: 1.33
D. None of the other choices is correct
E. mean: 1.39; standard deviation: 0.80

Suppose that a random variable X has the discrete uniform distribution on the integers 10, …, 20. Find P(X=7).

The on-line access computer service industry is growing at an extraordinary rate. Current estimates suggest that 20% of people with home-based computers have access to on-line services. Suppose that 15 people with home-based computers were randomly and independently sampled. What is the probability that at least 1 of those sampled have access to on-line services at home?

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.8^{15} = 0.9648$$

$$= \sum_{x=1}^{15} C_{15}^{x} 0.2^{x} 0.8^{15-x}$$

A random number generator generates random intergers from 1 to 10 following a uniform distribution. What is the probability of the random number generation a 7?

$$P(X=7)=\frac{1}{10-1+1}=\frac{1}{10}$$

A batch of 50 machined parts contains 5 that do not conform to customer requirements. Determine the range of the random variable that is the number of parts that do not conform to customer requirements in a sample of 8 parts selected without replacement from the batch.

a. {0, 1,…, 8}

b. None of the other choices is correct

C {0, 1,…, 50}

D {0, 1,…, 5}

On a multiple-choice test with 16 questions, each question has four possible answers, one of which is correct. For students who guess at all answers, find the standard deviation for the random variable X, the number of the correct answers

a. 3
b. None of the other choices is correct
C 2.316
D 1.732
E 1.677

$$\sigma = \sqrt{16 \times 0.25 \times 0.75} = 1.732$$

CHAPTER 4

Transportation officials tell us that 70% of drivers wear seat belts while driving. Find the probability that more than 579 drivers in a sample of 800 drivers wear seat belts.
Let P(Z<1.5)=0.0668; P(Z<0.25)=0.6

A. 0.0668

B. 0.9332

C. None of the other choices is correct

D. 0.4

E. 0.6

$$P(X \le x) = \mathrm{P}\left(X \le x + 0.5\right) \approx P(Z \le \frac{x + 0.5 - np}{\sqrt{np(1-p)}})$$

$$P(X \ge x) = \mathrm{P}\left(x - 0.5 \le X\right) \approx P(Z \ge \frac{x - 0.5 - np}{\sqrt{np(1-p)}})$$

$$X \sim \mathrm{B}(800, 0.7) \Rightarrow X \approx N(800*0.7; 800*0.7*0.3)$$

$$P(X > 579) = 1 - P(X \le 579) = 1 - P\left(Z \le \frac{579.5 - 800*0.7}{\sqrt{800*0.7*0.3}}\right)$$

$$= 1 - P(Z \le 1.5) = 1 - 0.0668 = 0.9332$$

The cumulative distribution function of the random variable X defined by

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-2x}, & x \geq 0. \end{cases}$$

What is $P[2 < X < \infty]$?

A. 0.3159

B. 0.5682

C. None of the other choices is correct

D. 0.2415

E. 0.0183

$$P(2 < X < \infty) = F(\infty) - F(2)$$
$$= 1 - e^{-\infty} - (1 - e^{-4}) = e^{-4}$$

Suppose X is a uniform random variable over the interval [40;50]. Find the probability that a randomly selected observation exceeds 43.

$$P(X > 43) = \int_{43}^{50} \frac{1}{10}\,dx = \frac{50-43}{10} = 0.7$$

A. 0.3

B. 0.1

C. None of the other choices is correct

D. 0.7

E. 0.9

The probability density function of the length of a hinge for fastening a door is $f(x) = 1$ for $74, x < 75$. Find $P(X < 74.3)$

$$P(X < 74.3) = 74.3 - 74 = 0.3$$

A. 0.81

B. 0.30

C. 0.36

D. 0.74

E. None of the other choices is correct

An airline knows from experience that the distribution of the number of suitcases that get lost each week on a certain route is approximately normal with $\mu = 15.5$ and $\sigma = 3.6$. What is the probability that during a given week the airline will lose less than 20 suitcases?

$$P(X < 20) = P\left(Z < \frac{20 - 15.5}{3.6}\right) = P(Z < 1.25) = 0.8944$$

A. 0.4040

B. 0.8944

C. 0.3944

D. 0.1056

E. None of the other choices is correct

Suppose X has an exponential distribution with λ = 2. Which in the following statements is TRUE?

A. P(X ≥ 2) = 0.0183

B. P(X ≤ 1) = 0.8647

C. P(X ≤ 0) = 0

D. All of the others

$$P(X \geq 2) = 1 - P(X < 2) = 1 - \int_0^2 2e^{-2x}dx = e^{-4} = 0.018$$

$$P(X \leq 1) = \int_0^1 2e^{-2x}dx = 0.8647$$

Let X be a random variable, its the probability density function is given by $f(x) = \frac{3}{2}\sqrt{x}$ when $0 < x < 1$. Determine $E[X^3]$.

A. 3/4

B. 2

C. 1

D. None of the other choices is correct

E. 1/3

$$E\left(X^3\right) = \int_0^1 x^3 \frac{3}{2}\sqrt{x}dx = \frac{1}{3}$$

If we know that the length of time it takes a college student to find a parking spot in the library parking lot follows a normal distribution with a mean of 3.5 minutes and a standard deviation of 1 minute, find the point in the distribution in which 75.8% of the college student exceed when trying to find a parking spot in the library parking lot.
Let P(Z<-0.7) = 0.242, P(Z<-0.1) = 0.460 and P(Z<0.7) = 0.758

a. 3.2 minutes
b. 4.2 minutes
C 3.4 minutes
D 2.8 minutes
E None of the other choices is true

$$P(X > x) = 1 - P(X < x) = 1 - P\left(Z < \frac{x - 3.5}{1}\right) = .0758$$

$$\Leftrightarrow P(Z < x - 3.5) = 0.242 \Leftrightarrow x - 3.5 = -0.7 \Rightarrow x = 2.8$$

Let X be a random variable that has the density function

$$f(x) = 2e^{-2x}, x > 0.$$

Denote F(x) the cumulative distribution function of X.
Calculate F(0.75).

a. None of the other choices is correct
b. 0.8543
C 0.7769
D 0.1457
E 0.2231

$$F(x) = \int_0^x 2e^{-2t}dt = -e^{-2t}\Big|_0^x = 1 - e^{-2x}$$

$$F(0.75) = 1 - e^{-2\times 0.75} = 0.7769$$

If we are using the normal approximation to determine the probability of at most 28 successes in a binomial distribution $P(X \leq 28)$ the normal distribution probability that is used to make the estimate is

(i) $P(X \leq 28.5)$

(ii) $P(X < 28)$

(iii) $P(X \leq 27.5)$

(iv) $P(X \leq 28)$

The weekly demand for Pepsi, in thousand of litters, from a local chain of efficiency stores, is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1) & \text{if } 1 < x < 2 \\ 0 & \text{if elsewhere} \end{cases}$$

Find the variance of X.

A catalog company that receives the majority of its orders by telephone conducted a study to determine how long customers were willing to wait on hold before ordering a product. The length of waiting time was found to be a random variable best approximated by an exponential distribution with a mean length of waiting time equal to 3 minutes (i.e. the mean number of calls answered in a minute is 1/3). What proportion of customers having to hold more than 4.5 minutes will hang up before placing an order?

a. None of the other choices is correct
b. 0.77687
C 0.22313
D 0.51342
E 0.48658

$$P\left(X > 4.5\right) = e^{-\frac{1}{3}\times 4.5} = 0.22313$$

CHAPTER 6,7

Suppose that $\mu = 16$ and $\sigma^2 = 20$ for a population. In a sample where $n = 100$ is randomly taken, what is the standard deviation for the sample mean?

A. 0.2

B. 0.16

C. 0.02

D. 0.45

E. None of the other choices is correct

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{20}}{10} = 0.45$$

At a computer manufacturing company, the actual size of computer chips is normally distributed with a mean of 1 centimeter and a standard deviation of 0.1 centimeter. A random sample of 12 computer chips is taken. What is the probability that the sample mean will be between 0.99 and 1.01 centimeters?
Let P(Z < 0.1) = 0.54, P(Z < 0.346) = 0.64, P(Z < -0.346) = 0.37.

A. 0.73 B. 0.63 C. 0.37 D. None of the other choices is correct E. 0.27

$$P\left(0.99 < \overline{X} < 1.01\right) = P\left(Z < \frac{1.01-1}{0.1/\sqrt{12}}\right) - P\left(Z < \frac{0.99-1}{0.1/\sqrt{12}}\right)$$

$$= P\left(Z < 0.346\right) - P\left(-0.346\right) = 0.27$$

The average score of all golfers for a particular course has a mean of 79 and a standard deviation of 5. Suppose 100 golfers played the course today. Find the probability that the average score of the 100 golfers exceeded 80.
Let P(Z<2)=0.9772, P(Z<0.2) = 0.5793

A. 0.0228
B. None of the other choices is correct
C. 0.4772
D. 0.1293
E. 0.3707

$$P(X>80)=1-P(X<80)=1-P\left(Z<\frac{80-79}{5/\sqrt{100}}\right)$$

$$=1-P(Z<2)=1-0.9772=0.0228$$

Each morning, a teacher quizzed his class with 20 geography questions. The class marked them together and everyone kept a record of their personal scores. As the year passed, each student tried to improve his or her quiz marks. Every day, Elliot recorded his quiz marks on a stem and leaf plot. This is what his marks looked like plotted out:

Stem	Leaf
1	3 4 5 5
2	0 2 3 4 2 4 5 6 4 4 8 7 9
3	1 1 1 0 1

A. 15
B. None of the other choices is correct
C. 24
D. 31
E. 31 and 24

Find the mode of the data

A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with a mean of 70 and standard deviation 12. Let $\bar{X}_1$ and $\bar{X}_2$ be the two sample means. Find $P(\bar{X}_1 - \bar{X}_2 > 5.5)$.

Let $P(Z \geq 0.11) = 0.5438$.

$X_1 \sim N(75,8), n_1 = 16;$

$X_2 \sim N(70,12), n_1 = 9;$

$$X_1 \sim N(75,8), n_1 = 16;\ X_2 \sim N(70,12), n_1 = 9;$$

$$E(\overline{X}_1 - \overline{X}_2) = E(\overline{X}_1) - E(\overline{X}2) = 75 - 70 = 5$$

$$V(\overline{X}_1 - \overline{X}_2) = V(\overline{X}_1) + V(\overline{X}2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

$$= \frac{8^2}{16} + \frac{12^2}{9} = 20$$

$$\overline{X}_1 - \overline{X}_2 \sim N(5,20)$$

$$P(\overline{X}_1 - \overline{X}_2 > 5.5) = P\left(Z > \frac{5.5-5}{\sqrt{20}}\right) = P(Z > 0.11) = 0.5438$$

Find the mode and the median of the sample 18, 19, 16, 21, 18, 19, 24, 15, 18

a. None of the other choices is correct

b. 18 and 19

C 19 and 18

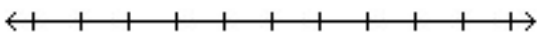
D 19 and 19

E 18 and 18

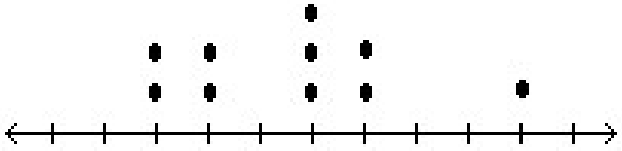
A store manager counts the number of customers who make a purchase in his store each day. The data are as follows.

10 11 8 14 7 10 10 11 8 7

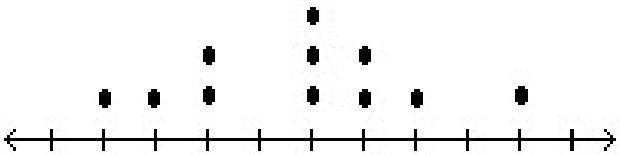
Construct the dot plot for the given data.



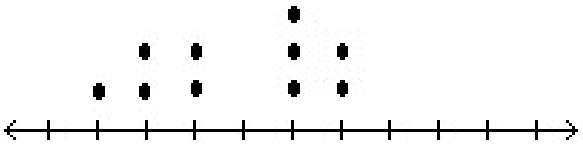
a.



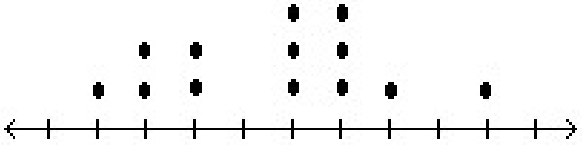
b.



c.



d.



A sociologist recently conducted a survey of senior citizens who have net worths too high to qualify for Medicaid but have no private health insurance. The ages of the 25 uninsured senior citizens were as follows:

69	74	67	77	87
75	62	90	66	91
70	93	77	63	82
64	69	82	71	74
61	88	76	65	83

Find the median of the observations.

The manager of an electrical supply store measured the diameters of the rolls of wire in the inventory. The diameters of the rolls (in m) are listed below:

0.165 0.114 0.503 0.392 0.579 0.311. Find the range of data.

Find the origin data from the sterm-and-leaf plot

Stem	Leaves
8	5 8
9	1 8
10	5 5

a. 81, 85, 81, 98, 108, 105

b. 85, 81, 88, 91, 101, 105

c. 85, 88, 91, 91, 105, 105

d. 85, 88, 91, 98, 105, 105

The amount of time required for an oil and filter change on an automobile is normally distributed with a mean of 48 minutes and a standard deviation of 10 minutes. A random sample of 36 cars is selected. What is the probability that the sample mean will be between 39 and 48 minutes?

a. 0.500

b. 0.8767

c. 0.3376

d. 0.8151

$$P(39 < \overline{X} < 48) = P(\overline{X} < 48) - P(\overline{X} \le 39)$$

$$= P\left(Z < \frac{48-48}{10/\sqrt{36}}\right) - P\left(Z \le \frac{39-48}{10/\sqrt{36}}\right) = 0.5$$

The time for a worker to assemble a component is normally distributed with mean 15 minutes and variance 4. Denote the mean assembly times of 16 day-shift workers and 9 night-shift workers by $\overline{X}$ and $\overline{Y}$, respectively. Assume that the assembly times of the workers are mutually independent. The distribution of $\overline{X}$ - $\overline{Y}$ is

a. standard normal.

b. normal with mean 15 and variance 5/6.

c. normal with mean 0 and standard deviation 5/6.

d. normal with mean 0 and standard deviation 25/72.

$$X_1 \sim N(15;4); X_2 \sim N(15;4), n_1 = 16; n_2 = 9$$

$$X_1 - X_2 \sim N(0;5/6)$$

CHAPTER 8

A laboratory tested 70 chicken eggs and found that the mean amount of cholesterol was 190 milligrams. Assume that the standard deviation is 15.1 milligrams. Construct a 95% lower confidence bound for the true mean cholesterol content of all such eggs.

A. 185.28

B. 187.03

C. None of the other choices is correct

D. 184.79

E. 186.46

$$\mu \geq \bar{x} - z_{\alpha}\frac{\sigma}{\sqrt{n}} = 190 - 1.645\frac{15.1}{\sqrt{70}} = 187.03$$

Let $z_{0.025} = 1.96,\ z_{0.05} = 1.645$

Construct a 98% confidence interval for the population mean. Assume the population has a normal distribution. A study of 14 bowlers showed that their average score was 192 with a standard deviation of 8

A. (222.3, 256.1)
B. (186.3, 197.7)
C. (115.4, 158.8)
D. (328.3, 386.9)
E. None of the other choices is correct

Let $t_{0.01,\,13} = 2.65$

$$\bar{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

$$192 - 2.65\frac{8}{\sqrt{14}} < \mu < 192 + 2.65\frac{8}{\sqrt{14}}$$

$$186.33 < \mu < 197.67$$

Of 1000 randomly selected cases of lung cancer, 731 resulted in death within 10 years. Calculate a 95% confidence interval on the death rate from lung cancer.

A. (0.703; 0.758)

B. (0.307; 0.751)

C. (0.707; 0.754)

D. None of the others

Let $z_{0.025} = 1.96,\ z_{0.05} = 1.65$

$$p = \frac{731}{1000} = 0.731$$

$$p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \le p \le p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

$$0.731 - 1.96\sqrt{\frac{0.731*0.269}{1000}} \le p \le 0.731 + 1.96\sqrt{\frac{0.731*0.269}{1000}}$$

$$0.7035 \le p \le 0.758$$

A group of 40 bowlers showed that their average score was 192 with a standard deviation of 8. Find the 95% confidence interval of the mean score of all bowlers.

A (186.5; 197.5)
B (188.5; 195.6)
C (187.3; 196.1)
D (189.5; 194.5)

Let

$z_{0.05} = 1.65, \quad z_{0.025} = 1.96,$

$t_{0.05,39} = 1.68, \quad t_{0.025,39} = 2.02.$

A Professor at Hanoi Medical University is interested in estimating the birth weight of infants. How large a sample must he select if he desires to be 99% confident that the true mean is within 0.1 kilograms of the sample mean? A past experience indicates that the standard deviation of the birth weights is known to be 0.7 kilograms. Let $z_{0.005} = 2.58$.

a. 327 b. 319 c. 301 d. 300

$$n = \left(\frac{z_{0.005} \times s}{E} \right)^2 = \left(\frac{2.58 \times 0.7}{0.1} \right)^2 = 327$$

A manufacturer of golf equipment wishes to estimate the number of left-handed golfers. How large a sample is needed in order to be 95% confident that the sample proportion will not differ from the true proportion by more than 2%? A previous study indicates that the proportion of left-handed golfers is 15%.

$$n = \left(\frac{z_{0.025}}{E} \right)^2 \times p(1-p) = \left(\frac{1.96}{0.02} \right)^2 0.15 \times 0.85 = 1225$$

The grade point averages for 11 randomly selected students in a statistics class are listed below.

2.4 3.2 1.8 1.9 2.9 4.0 3.3 0.9 3.6 0.8 2.2

What is the effect on the width of the confidence interval if the sample size is increased to 15?

a. The width remains the same.

b. It is impossible to tell without more information.

C The width decreases.

D The width increases.

A random sample of 10 parking meters in a beach community showed the following incomes for a day. Assume the incomes are normally distributed.
$3.60 $4.50 $2.80 $6.30 $2.60 $5.20 $6.75 $4.25 $8.00 $3.00
Find the 95% confidence interval for the true mean. |
Let $z_{0.025} = 1.96; t_{0.025,9} = 2.262$

a. ($3.39, $6.01)
b. ($1.35, $2.85)
C ($2.11, $5.34)
D ($4.81, $6.31)

$\bar{x} = 4.7,\ s = 1.83$

$$4.7 - 2.262\frac{1.83}{\sqrt{10}} < \mu < 4.7 + 2.262\frac{1.83}{\sqrt{10}}$$

$$3.39 < \mu < 6.01$$

CHAPTER 9

Given $H_0 : p = 0.85$ and $\alpha = 0.10$, which level of confidence should you use to test the claim?

A. 95%

B. 90%

C. None of the other choices is correct

D. 99%

E. 80%

Assume that IQ scores for a certain population are approximately N(μ; 100). To test the hypothesis H_0: $\mu = 110$ against H_1 : $\mu > 110$, we take a random sample of size n = 16 from this population and observe $\bar{x} = 113.5$.
Do we accept or reject H_0 at the 5% level?

Let $z_{0.05} = 1.645$

$$z_0 = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{113.5 - 110}{10/4} = 1.4$$

A. Fail to reject Ho

B. Reject Ho

$$z_0 < z_{0.05} \Rightarrow A$$

Medicare would like to test the hypothesis that the average monthly rate for one-bedroom assisted-living facility is equal to $3,300. A random sample of 12 assisted-living facilities had an average rate of $3,690 per month and a standard of $530. It is believed that the monthly rate for one-bedroom assisted-living facility is normally distributed. Use the signinicance level of 0.05 for this hypothesis test, what is the value of the test statistic?

A. -1.37

B. None of the other choices is true

C. -2.16

D. 2.55

E. 2.21

$$t_0 = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3690 - 3300}{530 / \sqrt{12}} = 2.55$$

Trang Tien is a producer of ice cream and would like to test the hypothesis that the average consumes more than 17 ounces of ice cream per month. A random sample of 15 Vietnamese people was found to consume an average of 18.2 ounces of ice cream last month. The standard deviation for this sample was 3.9 ounces. What is the test statistic for this hypothesis test?

A. None of the other choices is true

B. 1.19

C. 3.01

D. 1.83

E. 2.49

$$t_0 = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{18.2 - 17}{3.9 / \sqrt{15}} = 1.19$$

A soft drink company has a filling machine that can be set at different levels to produce different average fill amounts. The company sets the machine to provide a mean fill of 12 ounces. The standard deviation on the machine is known to be 0.20 ounces. Assuming t hat the hypothesis test is to be performed using a random sample of $n = 100$ cans, which of the following would be the correct formulation of the null and alternative?

(ii) $H_0 : \mu = 12$ ounces
$H_1 : \mu \neq 12$ ounces

You wish to test the claim that $\mu \leq 38$ at a level of significance of $\alpha = 0.01$ and are given sample statistics $n = 43, s = 4.7, \bar{x} = 39.8$. Compute the value of the test statistic. Round your answer to two decimal places.

A 3.51
B 2.65
C None of the other choices is true
D 2.51
E 1.96

$$z_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{39.8 - 38}{4.7 / \sqrt{43}} = 2.51$$

Find the test statistic t_0 for a sample with $n = 17, \bar{x} = 17.7, s = 2.4$ and $\alpha = 0.01$ if $H_1 : \mu \neq 17.9$. Round your answer to three decimal places.

A. -0.008
B. None of the other choices is correct
C -0.344
D -0.825
E -0.381

$$t_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{17.7 - 17.9}{2.4 / \sqrt{17}} = -0.344$$

A telephone company claims that 25% of its customers have at least two telephone lines. The company selects a random sample of 500 customers and finds that 108 have two or more telephone lines. At $\alpha = 0.05$, compute the value of the test statistic to test the company's claim.

a. 1.76
b. -2.33
C -1.76
D 2.33

$$z_0 = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{\frac{108}{500} - 0.25}{\sqrt{\frac{0.25 \times 0.75}{500}}} = -1.76$$

Expedia would like to test the hypothesis that the proportion of united airlines flights that arrive on time is less than 0.80. A random sample of 110 united airline flights found that 82 arrived on-time. Expedia would like to set the significant level to be 0.02. the test statistic for this hypothesis test would be_____

A. -2.77
B. -1.87
C -2.3
D -1.43

$$p = \frac{82}{110} = 0.745; \qquad z_0 = \frac{p - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.745 - 0.8}{\sqrt{\frac{0.8 * 0.2}{110}}} = -1.43$$

An article stated that students in FPT university system take an average of 4.5 years to finish their undergraduate degrees. Suppose you believe that the average time is longer. You conduct a survey of 49 students and obtain a sample mean of 5.1 with a sample standard deviation of 1.2. Assume that time to finish their undergraduate degrees is normally distributed. Calculate the value of the test statistic and the critical values for this test statistic. Use a significance level of 0.05.

Let

$$z_{0.05} = 1.645, \quad z_{0.025} = 1.96, \quad t_{0.05,48} = 2.011, \quad t_{0.025,48} = 2.314$$

A. Test statistic: 3.5 and critical value:1.645
B. None of the other choices is correct
C Test statistic: 3.5 and critical value: 2.011
D Test statistic: 3.5 and critical value: 1.96
E Test statistic: -3.5 and critical value: -2.314

$$n = 49, \bar{x} = 5.1, s = 1.2, \mu_0 = 4.5$$

$$H_1 : \mu > 4.5$$

$$z_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{5.1 - 4.5}{1.2 / \sqrt{49}} = 3.5$$

$$z_{0.05} = 1.645$$

A bearing used in an automotive application is supposed to have a norminal inside diameter of 1.5 inches. A random sample of 36 bearing is selected and the average inside diameter of these bearing is 1.4895 inches. Bearing diameter is known to be normally distributed with standard deviation $\sigma = 0.02$ inch. Test the hypotheses $H_0 : \mu = 1.5$ versus $H_1 : \mu \neq 1.5$ using $\alpha = 0.01$

Let

$$z_{0.01} = 2.326 \ , z_{0.005} = 2.576$$

$$n = 36, \bar{x} = 1.4895, \sigma = 0.02, \mu_0 = 1.5, \alpha = 0.01$$

$$z_{0.05} = 2.576$$

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{1.4895 - 1.5}{0.02 / \sqrt{36}} = -3.15$$

Reject H0

CHAPTER 10

Assume that the two population variances are equal and that they have been randomly selected.

Independent samples from two different populations yield the following data.

$\overline{x}_1 = 383; \overline{x}_2 = 448, s_1 = 44; s_2 = 18.$

$$\left(\overline{x_1} - \overline{x_2} \pm z_{0.025} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right) = (-72; -58)$$

The sample size is 177 for both samples. Find the 95 percent confidence interval for $\mu_1 - \mu_2$

a. (-72; -58

b. (-66; -64)

C (-79; -51)

D (-70; -60)

E None of the other choices is correct.

$$\overline{x_1} - \overline{x_2} = 383 - 448 = -65, z_{0.025} = 1.96$$

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{44^2}{177} + \frac{18^2}{177}} = 3.573$$

$$\overline{x_1} - \overline{x_2} = 383 - 448 = -65, z_{0.025} = 1.96$$

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{44^2}{177} + \frac{18^2}{177}} = 3.573$$

$$\left(\overline{x_1} - \overline{x_2} \pm z_{0.025} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right) = (-72; -58)$$

Compute the test statistic used to test the null hypothesis that $p_1 = p_2$.

$n_1 = 187;\ n_2 = 177$

$x_1 = 67\ ;\ x_2 = 62$

A. 2.200
B. 0.364
C 0.160
D 4.086

$$p_1 = \frac{67}{187} = 0.358;\ p_2 = \frac{62}{177} = 0.350;\ P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{129}{364} = 0.354$$

$$Z_0 = \frac{P_1 - P_2}{\sqrt{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}},$$

The concentration of active ingredient in a liquid laundry detergent is thought to be affected by the type of catalyst used in the process. The standard deviation of active concentration is known to be 3 grams per liter regardless of the catalyst type. Ten observations on concentration are taken with each catalyst, and the data follow:

Catalyst 1: 57.9, 66.2, 65.4, 65.4, 65.2, 62.6, 67.6, 63.7, 67.2, 71.0

Catalyst 2: 66.4, 71.7, 70.3, 69.3, 64.8, 69.6, 68.6, 69.4, 65.3, 68.8

Find a 95% confidence interval on the difference in mean active concentrations for the two catalysts.

A. (0.55; 3.82)
B. (-6.75; 0.31)
C (-5.83; -0.57)
D None of the other choices is correct

Find the test statistic to test the claim that $\mu_1 = \mu_2$. Two samples are randomly selected and come from populations that are normal with the same variance. The sample statistics are given below.

$n_1 = 25, \bar{x}_1 = 20, s_1 = 1.5$
$n_2 = 30, \bar{x}_2 = 18, s_2 = 1.9$

a. 4.27
b. -4.27
C 1.63
D -1.63
E None of the other choices is correct.

Compute the test statistic used to test the null hypothesis that $p_1 = p_2$.

A random sampling of sixty pitchers from the National League and fifty-two pitchers from the American League showed that 16 National and 11 American League pitchers had E.R.A's below 3.5.

A. 12.635
B. 0.68
C 0.884
D 10.184
E None of the other choices is correct.

Consider the hypothesis test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$ with known variances $\sigma_1 = 10$ and $\sigma_2 = 5$. Suppose that sample sizes $n_1 = 10$ and $n_2 = 15$ and that $\bar{x}_1 = 24.5$ and $\bar{x}_2 = 21.3$. Use $\alpha = 0.01$.

Calculate the value of the test statistic.

a. 0.937
b. 1.37
c. 1.37
d. 0.937

CHAPTER 11

Over a period of one year, a greengrocer sells tomatoes at six different prices (x pance per kilogram). He calculates the average number of kilograms, y, sold per day at each of the six different prices. From these data the following are calculated

$$\sum x_i = 200, \sum y_i = 436,$$
$$\sum x_i y_i = 12515, \sum x_i^2 = 7250, \sum y_i^2 = 39234, n = 6.$$

Estimate the correlation coefficient.

A. 0.692

B. 0.962

C. -0.962

D. None of the other choices is correct

E. -0.692

$$R = \frac{S_{XY}}{\sqrt{S_{XX} SS_T}} = \frac{\sum xy - \sum x \sum y / n}{\sqrt{\left(\sum x^2 - \left(\sum x\right)^2 / n\right)\left(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2\right)}}$$

The residuals represent

A. None of the other choices is correct

B. the difference between the actual Y values and the predicted Y values.

C. the difference between the actual Y values and the mean of Y.

D. the square root of the slope.

E. the predicted value of Y for the average X value.

The height y and base diameter x of five tree of a certain variety produced the following data

x	2	2	3	5
y	31	36	94	127

Find the equation of the estimated regression line $\hat{y} = a + bx$ of y on x.

A. -21.5x + 31.17
B. 21.5+31.17x
C. None of the other choices is correct
D. -21.5+31.17x
E. 21.5x+31.17

Use the following table to find the value of the estimated variance, $\hat{\sigma}^2$, for the simple regression model.

SUMMARY OUTPUT			
Regression Statistics			
Multiple R	0.756604788		
R Square	0.572450805		
Adjusted R Square	0.429934407		
Standard Error			
Observations	5		
ANOVA			
	df	*SS*	*MS*
Regression	1	9.846153846	9.846153846
Residual			
Total	4	17.2	
	Coefficients	*Standard Error*	*t Stat*
Intercept	1.138461538	1.413767192	0.805268042
X Variable 1	0.615384615	0.307050613	2.004179733

$$SSE = 17.2 - 9.8 = 7.4$$

$$\sigma^2 = SSE/(n-2) = 7.4/3 = 2.45$$

Given the size of a human's brain, x, and their score on an IQ test, y, would you expect a positive correlation, a negative correlation, or no correlation?

A. Positive correlation

B. Negative correlation

C no correlation

To get the best deal on a microwave oven, Amanda called six appliance stores and asked the cost and volume of several models. The prices and volumes she was quoted are listed below. Compute the correlation coefficient.

Volume	1.9	2.2	0.5	1.7	1.5	0.5
Price	183	121	49	208	278	67

a. 0.757 b. -0.595 c. -0.757 d. 0.595

For the data in the table below, find the equation of the regression line of y on x.

X: 0 1 2 1

Y: 0 0 1 3

A. $y = x$

B. $y = 0.5x + 0.5$

C $y = 0x + 1$

D $y = 2.4 - x$

E None of the other choices is true

The weight and systolic blood pressure of 4 randomly selected males in the age group 25 to 30 are shown in the following table. Assume that weight and blood pressure are jointly normally distributed.

Weight, x	50	62	67	55
Pressure, y	90	110	100	90

What is the value of the test statistic for testing $H_0 : \beta_1 = 0$

a. 1.46 b. 1.19 c. 0.36 d. 0.92

$$\beta_0 = 43.846; \beta_1 = 0.917;$$

$$S_{xx} = 13858 - \frac{234^2}{4} = 169; S_{xy} = 22970 - \frac{234*390}{4} = 155$$

$$SST = 38300 - \frac{390^2}{4} = 275; SSE = 275 - 0.917*155 = 132.865$$

$$\sigma^2 = \frac{132.865}{2} = 66.4325$$

$$t_0 = \frac{\beta_1 - \beta_{1,0}}{\sqrt{\sigma^2 / S_{xx}}} = \frac{0.917}{\sqrt{66.4325/169}} = 1.46$$